№ 11

На семинаре: $\tilde{\theta}_1 = 2\bar{x}$ $\quad \tilde{\theta}_2 = x_{min}$ $\quad \tilde{\theta}_3 = x_{max}$ $\quad \tilde{\theta}_4 = x_1 + \frac{1}{(n-1)}\sum_{i=2}^{n} x_i$

несм.	✓	✗	✗	✓
сост	✓	✗	?	✗

Подарок:

Док-ть: 1) по опр. $\tilde{\theta}_3' = \frac{n+1}{n} x_{max}$ – сост.

2) $\tilde{\theta}_3$ – сост

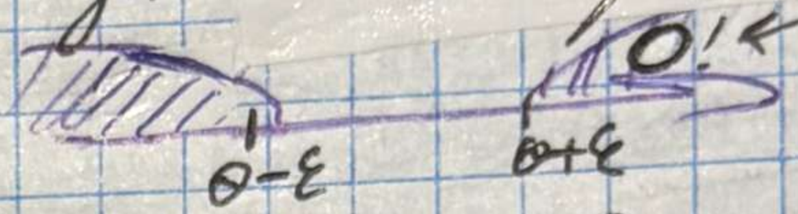
2) Оцн. сост. $\forall \theta>0\ \forall \varepsilon>0\ P(|\tilde\theta-\theta|\geq\varepsilon)\xrightarrow[n\to\infty]{}0$

$\tilde\theta_3 = x_{max}$ $\qquad \Psi(x)=\left(\frac{x}{\theta}\right)^n\{(0;\theta)\}$ $\qquad \psi(x)=\Psi'(x)=n\left(\frac{x}{\theta}\right)^{n-1}\frac{1}{\theta}\{(0;\theta)\}$

$M\tilde\theta_3 = Mx_{max} = \frac{n}{n+1}\theta$ – смещ. оц. $\Rightarrow$ не может быть дост.

Дл., поэтому найдём док-ть по опр.: т.к. $\xi\sim R(0,\theta)$ то $P(x_{max}\geq\theta+\varepsilon)=0$

$P(|\tilde\theta_3-\theta|\geq\varepsilon)\xrightarrow[n\to\infty]{?}0$



$P(|\tilde\theta_3-\theta|\geq\varepsilon)=P(\tilde\theta_3\leq\theta-\varepsilon)=P(x_{max}\leq\theta-\varepsilon)=$

$=\Psi(\theta-\varepsilon)=\left(\frac{\theta-\varepsilon}{\theta}\right)^n=\left(1-\frac{\varepsilon}{\theta}\right)^n\xrightarrow[n\to\infty]{}0$

т.к. $0<\varepsilon<\theta$ $\frac{\varepsilon}{\theta}<1\Rightarrow\xrightarrow[n\to\infty]{}0$

а $\varepsilon\geq\theta$ не будет, всё лежит в $[0;\theta]$

$\Rightarrow\tilde\theta_3$ – сост. ✓

1) $\tilde\theta_3' = \frac{n+1}{n}x_{max}$

Докажем по опр. $\forall\varepsilon>0\ \forall\theta>0\ P(|\tilde\theta_3'-\theta|\geq\varepsilon)\xrightarrow[n\to\infty]{}0$

$P(|\tilde\theta_3'-\theta|\geq\varepsilon)=P(\tilde\theta_3'\leq\theta-\varepsilon)+P(\tilde\theta_3'\geq\theta+\varepsilon)=P\left(\frac{n+1}{n}x_{max}\leq\theta-\varepsilon\right)+P\left(\frac{n+1}{n}x_{max}\geq\theta+\varepsilon\right)=P\left(x_{max}\leq\frac{n(\theta-\varepsilon)}{n+1}\right)+P\left(x_{max}\geq\frac{n(\theta+\varepsilon)}{n+1}\right)$

$=\underbrace{\Psi\left(\frac{n(\theta-\varepsilon)}{n+1}\right)}_{\Psi^1}+1-\underbrace{\Psi\left(\frac{n(\theta+\varepsilon)}{n+1}\right)}_{\Psi^2}$ ⓥ

$\Psi^1=\left(\frac{n(\theta-\varepsilon)}{(n+1)\theta}\right)^n=\left(\frac{n}{n+1}-\frac{n\varepsilon}{(n+1)\theta}\right)^n=\overbrace{\left(\frac{n}{n+1}\right)^n}^{\to\frac{1}{e}}\overbrace{\left(1-\frac{\varepsilon}{\theta}\right)^n}^{\to 0}\xrightarrow[n\to\infty]{}0$

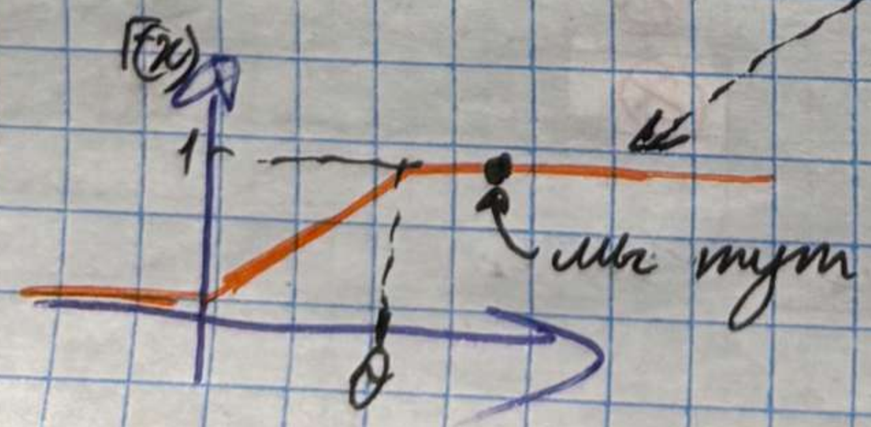
$\Psi^2=\left(\frac{n(\theta+\varepsilon)}{\theta(n+1)}\right)^n$ $\qquad \frac{n(\theta+\varepsilon)}{(n+1)}\leq\theta\Rightarrow n\theta+n\varepsilon\leq n\theta+\theta\Rightarrow n\leq\frac{\theta}{\varepsilon}\Rightarrow$

$\forall\varepsilon>0\ \exists N\in\mathbb{N}\ \forall n\geq N\hookrightarrow\frac{n(\theta+\varepsilon)}{n+1}>\theta\Rightarrow\Psi^2=(F(x))^n=1^n$

$\Rightarrow\Psi^2\xrightarrow[n\to\infty]{}1$

$\Rightarrow$ ⓥ $=0+1-1=0$

$\Rightarrow\tilde\theta_3'$ – сост. ✓



ч.т.д.